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Digital Signal Processing

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Analog domain frequency conversion

- Question
- Analog domain frequency conversion
- Prototype low-pass to high-pass conversion
- Prototype low-pass to band-pass conversion
- Prototype low pass to band resistance conversion
- Analog high-pass filter design
- Analog bandpass filter design
- Analog band-stop filter design

Problem

BW, CB, elliptic analog low-pass filter design template

How to design analog high-pass, band-pass, band-stop filters?

ω_p, ω_s —(Analog frequency / Rate conversion)→ $\bar{\omega}_p, \bar{\omega}_s$ —(Design prototype / Low-pass filter)→ $H_L(\bar{s})$ —(Complex frequency / convert)→ $H(s)$

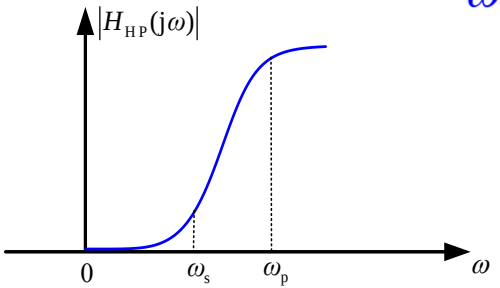
Analog domain frequency conversion

❖ Prototype low-pass to high-pass conversion

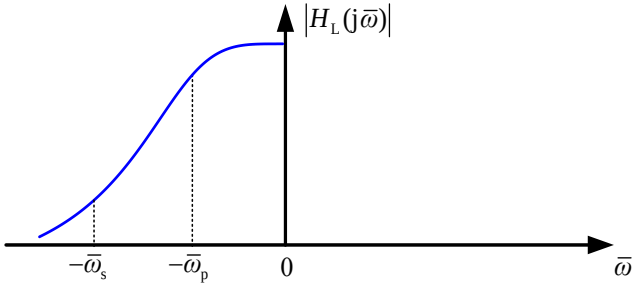
$$H_{\mathrm{HP}}(s) = H_{\mathrm{L}}(\bar{s})\big|_{\bar{s}=1/s}$$

$\bar{s} = \mathrm{j}\bar{\omega}, s = \mathrm{j}\omega$ $\quad \bar{s} = 1/s$ The frequency mapping relationship is substituted

$$\bar{\omega} = -1/\omega$$



Amplitude-frequency response of high-pass filter



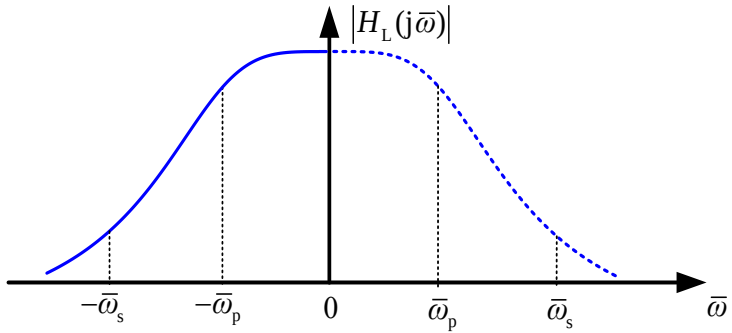
Amplitude-frequency response of prototype low-pass filter

Analog domain frequency conversion

❖ Prototype low-pass to high-pass conversion

Because the system function of the physically realizable filter should be real coefficient, its amplitude-frequency response has even symmetry. According to the symmetry characteristic of the amplitude-frequency response of the prototype low-pass filter, the positive frequency partial symmetric amplitude-frequency response is constructed, as shown by the dotted line in the figure. Therefore, the design index of the prototype low-pass filter can be obtained as follows:

$$\bar{\omega}_{\mathrm{p}} = 1/\omega_{\mathrm{p}}; \qquad \bar{\omega}_{\mathrm{s}} = 1/\omega_{\mathrm{s}}$$



Analog high-pass filter design

❖ Design steps of analog high-pass filter

(1) The high-pass design index is converted to the prototype low-pass design index by frequency mapping

$$\bar{\omega}_{\mathrm{p}} = 1/\omega_{\mathrm{p}} \qquad \bar{\omega}_{\mathrm{s}} = 1/\omega_{\mathrm{s}}$$

(2) Conversion of prototype low-pass to Qualcomm $H_{\mathrm{HP}}(s)$ by complex frequency conversion

$$H_{\mathrm{HP}}(s) = H_{\mathrm{L}}(\bar{s})\Big|_{\bar{s}=\frac{1}{s}}$$

[Example] Try to design a BW analog high-pass filter that meets the following criteria f_p=5kHz, f_s=1kHz, $A_p \le 1$dB, $A_s \ge 40$dB

Solution: (1) The high-pass filter design index is converted to the prototype low-pass design index

$$\bar{\omega}_p = 1/(2\pi \times 5000),\ \bar{\omega}_s = 1/(2\pi \times 1000)\ ,\ A_p \le 1\text{dB},\ A_s \ge 40\text{dB}$$

(2) Design BW prototype low-pass filter

$$N \ge \frac{\lg\left(\dfrac{10^{0.1A_s}-1}{10^{0.1A_p}-1}\right)}{2\lg(\bar{\omega}_s/\bar{\omega}_p)} = 3.28 \qquad N=4$$

$$\omega_c = \frac{\bar{\omega}_s}{(10^{0.1A_s}-1)^{1/(2N)}} = 5.033\times 10^{-5}$$

[Example] Try to design a BW analog high-pass filter that meets the following criteria f_p=5kHz, f_s=1kHz, $A_p \le$1dB, $A_s \ge$40dB

The system function of BW prototype low-pass filter is

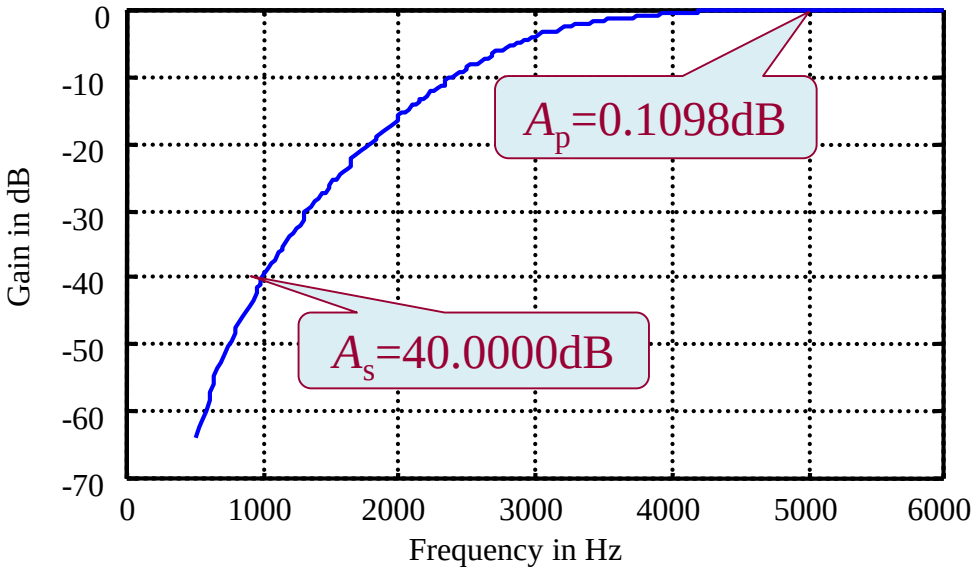
$$H_L(\bar{s}) = \frac{1}{[(\bar{s}/\omega_c)^2 + 0.7654\bar{s}/\omega_c + 1][(\bar{s}/\omega_c)^2 + 1.8478\bar{s}/\omega_c + 1]}$$

$$= \frac{6.4166\times10^{-18}}{\bar{s}^4 + 1.3152\times10^{-4}\bar{s}^3 + 8.6485\times10^{-9}\bar{s}^2 + 3.3315\times10^{-13}\bar{s} + 6.4166\times10^{-18}}$$

(3) The high-pass filter system function is obtained by complex frequency conversion $H_{HP}(s)=H_L(1/s)$

$$H_{HP}(s) = H_L(\bar{s})\big|_{\bar{s}=\frac{1}{s}} = \frac{s^4}{s^4 + 5.192\times10^4 s^3 + 1.3478\times10^9 s^2 + 2.0497\times10^{13} s + 1.5585\times10^{17}}$$

[Example] Try to design a BW analog high-pass filter that meets the following criteria f_p=5kHz, f_s=1kHz, $A_p \leq$1dB, $A_s \geq$40dB



Analog domain frequency conversion

❖ Prototype low-pass to band-pass conversion

$$H_{\mathrm{BP}}(s) = H_{\mathrm{L}}(\bar{s})\Big|_{\bar{s}=\frac{s^2+\omega_0^2}{Bs}}$$

Pass-band width $B = \omega_{\mathrm{p}2} - \omega_{\mathrm{p}1}$ Center frequency: $\omega_0 = \sqrt{\omega_{\mathrm{p}1}\omega_{\mathrm{p}2}}$

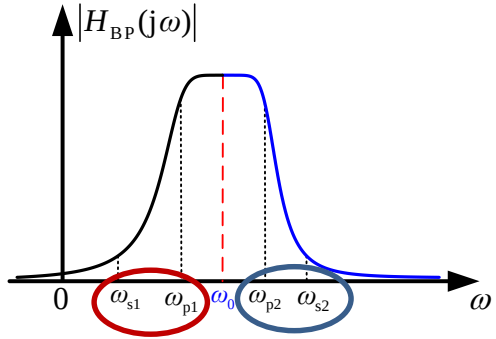
$$\bar{s} = \mathrm{j}\bar{\omega}, s = \mathrm{j}\omega \quad \bar{s} = \frac{s^2 + \omega_0^2}{Bs}$$

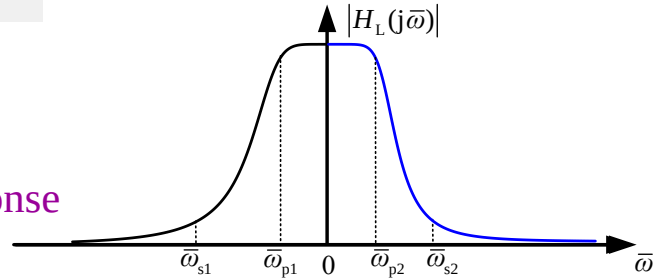
Medium frequency mapping relationship: $\bar{\omega} = \dfrac{\omega^2 - \omega_0^2}{B\omega}$

Analog domain frequency conversion

❖ Prototype low-pass to band-pass conversion

$$\bar{\omega} = \frac{\omega^2 - \omega_0^2}{B\omega}$$





The amplitude-frequency response is asymmetrical

$$\bar{\omega}_{p1} = \frac{\omega_{p1}^2 - \omega_{p1}\omega_{p2}}{(\omega_{p2} - \omega_{p1})\omega_{p1}} = -1$$

$$\bar{\omega}_{p2} = \frac{\omega_{p2}^2 - \omega_{p1}\omega_{p2}}{(\omega_{p2} - \omega_{p1})\omega_{p2}} = 1$$

$\Rightarrow \bar{\omega}_p = 1$

$$\bar{\omega}_{s1} = \frac{\omega_{s1}^2 - \omega_0^2}{B\omega_{s1}}$$

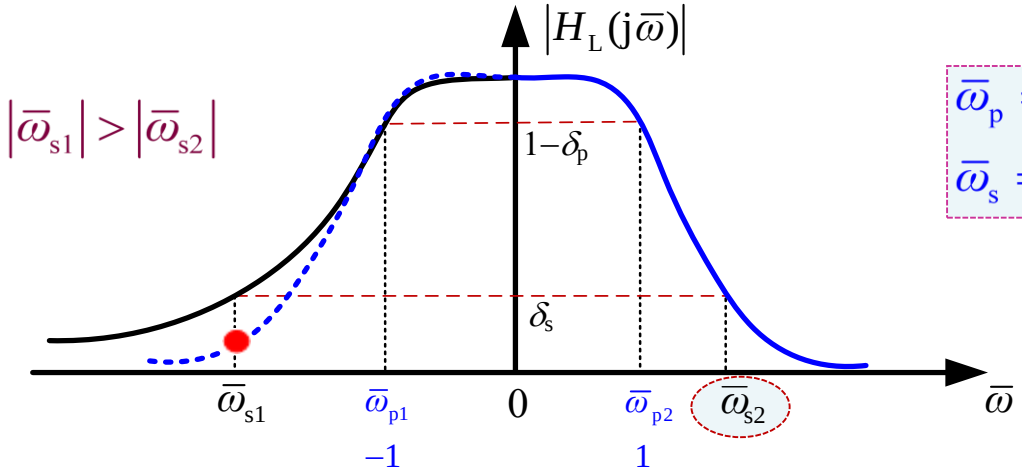
$$\bar{\omega}_{s2} = \frac{\omega_{s2}^2 - \omega_0^2}{B\omega_{s2}}$$

$\Rightarrow \bar{\omega}_s = ?$

Analog domain frequency conversion

❖ Prototype low-pass to band-pass conversion

$$\bar{\omega} = \frac{\omega^2 - \omega_0^2}{B\omega}$$



$$\bar{\omega}_p = |\omega_{p1}| = |\omega_{p2}| = 1$$

$$\bar{\omega}_s = \min\{|\bar{\omega}_{s1}|, |\bar{\omega}_{s2}|\}$$

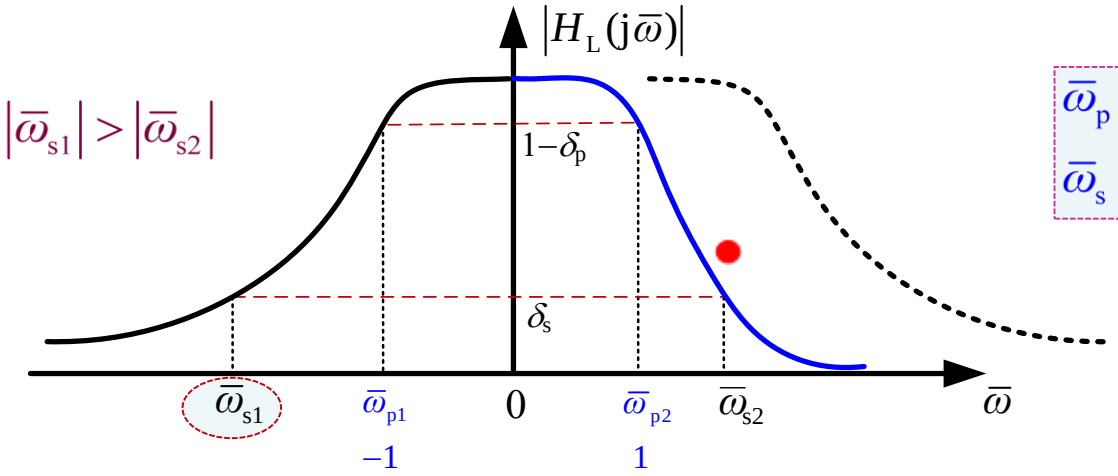
$$\bar{\omega}_s = \min\{|\bar{\omega}_{s1}|, |\bar{\omega}_{s2}|\}$$

Construct prototype low-pass filter ✓

Analog domain frequency conversion

❖ Prototype low-pass to band-pass conversion

$$\bar{\omega} = \frac{\omega^2 - \omega_0^2}{B\omega}$$



$$\bar{\omega}_p = |\omega_{p1}| = |\omega_{p2}| = 1$$

$$\bar{\omega}_s = \min\{|\bar{\omega}_{s1}|,\ |\bar{\omega}_{s2}|\}$$

$$\bar{\omega}_s = \max\{|\bar{\omega}_{s1}|,\ |\bar{\omega}_{s2}|\}$$

Construct prototype low-pass filter



Analog bandpass filter design

❖ Design steps of analog bandpass filter

(1) The parameters in the conversion formula are determined by the passband intercept frequency of the bandpass filter

$$B = \omega_{p2} - \omega_{p1} \qquad \omega_0^2 = \omega_{p1}\omega_{p2}$$

(2) The on-band and stop-band truncation frequency of prototype low-pass filter is determined ($\bar{\omega}_p$, $\bar{\omega}_s$)

$$\bar{\omega}_{s1} = \frac{\omega_{s1}^2 - \omega_0^2}{B\omega_{s1}} \qquad \bar{\omega}_{s2} = \frac{\omega_{s2}^2 - \omega_0^2}{B\omega_{s2}} \qquad \bar{\omega} = \frac{\omega^2 - \omega_0^2}{B\omega}$$

$$\bar{\omega}_p = |\omega_{p1}| = |\omega_{p2}| = 1$$

$$\bar{\omega}_s = \min\{|\bar{\omega}_{s1}|, |\bar{\omega}_{s2}|\}$$

Analog bandpass filter design

❖ Design steps of analog bandpass filter

(3) The design passband intercept frequency is $\bar{\omega}_p$ =1 (rad/s), stopband intercept frequency is $\bar{\omega}_s$. A prototype low-pass filter with passband attenuation of A_p dB and stopband attenuation of A_s dB

(4) Convert prototype low-pass filter to bandpass filter $H_{BP}(s)$

$$H_{BP}(s) = H_L(\bar{s})\Big|_{\bar{s}=\frac{s^2+\omega_0^2}{Bs}}$$

[Example] Try to design a BW analog bandpass filter that meets the following criteria ω_{p1}=6 rad/s, ω_{p2}=8 rad/s, $A_p \le 1$dB, ω_{s1}=4 rad/s, ω_{s2}=11 rad/s, $A_s \ge 32$dB

Solution: (1) The parameters in the conversion formula are determined by the passband intercept frequency of the bandpass filter

$$B = \omega_{p2} - \omega_{p1} = 2 \qquad \omega_0^2 = \omega_{p1}\omega_{p2} = 48$$

(2) The on-band and stop-band truncation frequency of prototype low-pass filter is determined ($\bar{\omega}_p$, $\bar{\omega}_s$)

$$\bar{\omega}_{s1} = \frac{\omega_{s1}^2 - \omega_0^{\,2}}{B\omega_{s1}} = -4 \qquad \bar{\omega}_{s2} = \frac{\omega_{s2}^2 - \omega_0^{\,2}}{B\omega_{s2}} = 3.3182 \qquad \bar{\omega} = \frac{\omega^2 - \omega_0^2}{B\omega}$$

so $\bar{\omega}_s = \min\{|\bar{\omega}_{s1}|, |\bar{\omega}_{s2}|\} = 3.3182$ $\qquad \bar{\omega}_p = 1$

[Example] Try to design a BW analog bandpass filter that meets the following criteria ω_{p1}=6 rad/s, ω_{p2}=8 rad/s, $A_p \le 1$dB, ω_{s1}=4 rad/s, ω_{s2}=11 rad/s, $A_s \ge 32$dB

Solution: (3) Design a prototype low-pass filter that meets the following criteria

$$\bar{\omega}_p = 1,\ A_p = 1\text{dB},\ \bar{\omega}_s = 3.3182,\ A_s = 32\text{dB}$$

$$N \ge \frac{\lg\left(\dfrac{10^{0.1A_s}-1}{10^{0.1A_p}-1}\right)}{2\lg(\bar{\omega}_s/\bar{\omega}_p)} = 3.6346 \quad N=4 \qquad \omega_c = \frac{\bar{\omega}_s}{(10^{0.1A_s}-1)^{1/(2N)}} = 1.3211$$

$$H_L(\bar{s}) = \frac{1}{[(\bar{s}/\omega_c)^2 + 0.7654\bar{s}/\omega_c + 1][(\bar{s}/\omega_c)^2 + 1.8478\bar{s}/\omega_c + 1]}$$

[Example] Try to design a BW analog bandpass filter that meets the following criteria ω_{p1}=6 rad/s, ω_{p2}=8 rad/s, $A_p \le 1$dB, ω_{s1}=4 rad/s, ω_{s2}=11 rad/s, $A_s \ge 32$dB

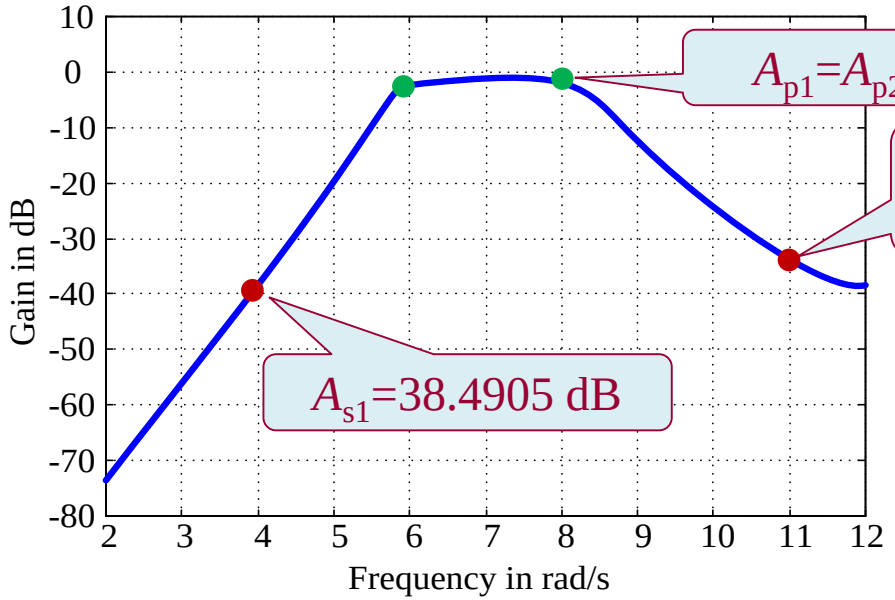
Solution: (4) Convert prototype low-pass filter to bandpass filter $H_{BP}(s)$

$$H_{BP}(s) = H_L(\bar{s})\Big|_{\bar{s}=\frac{s^2+\omega_0^2}{Bs}}$$

$$= \frac{48.7372s^4}{(s^4 + 2s^3 + 0.103\times10^3 s^2 + 0.0971\times10^3 s + 2.304\times10^3)} \times$$

$$\frac{1}{(s^4 + 0.0049\times10^3 s^3 + 0.103\times10^3 s^2 + 0.2343\times10^3 s + 2.304\times10^3)}$$

[Example] Try to design a BW analog bandpass filter that meets the following criteria ω_{p1}=6 rad/s, ω_{p2}=8 rad/s, $A_p \le 1$dB, ω_{s1}=4 rad/s, ω_{s2}=11 rad/s, $A_s \ge 32$dB



$$\bar{\omega}_{s1} = -4, \quad \bar{\omega}_{s2} = 3.3182$$

$$\bar{\omega}_s = \min\{|\bar{\omega}_{s1}|, |\bar{\omega}_{s2}|\} = 3.3182$$

$$\bar{\omega}_p = 1$$

Analog domain frequency conversion

❖ Prototype low pass to band resistance conversion

$$H_{\mathrm{BS}}(s)=H_{\mathrm{L}}(\bar{s})\Big|_{\bar{s}=\frac{Bs}{s^2+\omega_0^2}}$$

Stopband width: $B=\omega_{\mathrm{s}2}-\omega_{\mathrm{s}1}$ Center frequency: $\omega_0=\sqrt{\omega_{\mathrm{s}1}\omega_{\mathrm{s}2}}$

$$\bar{s}=\mathrm{j}\bar{\omega},s=\mathrm{j}\omega \quad \bar{s}=\frac{Bs}{s^2+\omega_0^2}$$

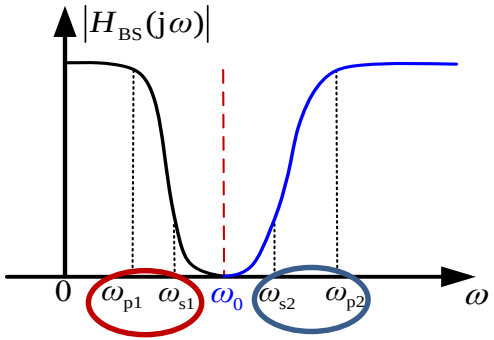
The frequency mapping relationship is obtained: $\bar{\omega}=\dfrac{B\omega}{-\omega^2+\omega_0^2}$

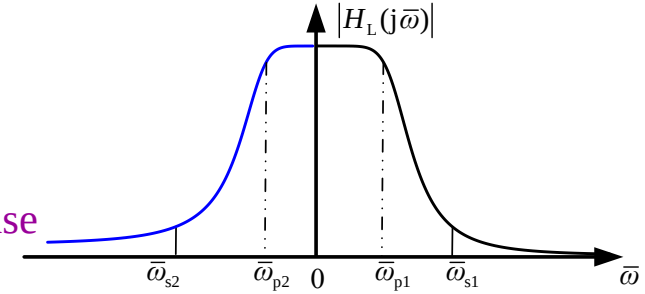
Analog domain frequency conversion

❖ Prototype low pass to band resistance conversion

$$\omega_0 = \sqrt{\omega_{s1}\omega_{s2}}$$

$$\bar{\omega} = \frac{B\omega}{-\omega^2 + \omega_0^2}$$





The amplitude-frequency response is asymmetrical

$$\bar{\omega}_{s1} = \frac{(\omega_{s2} - \omega_{s1})\omega_{s1}}{-\omega_{s1}^2 + \omega_{s1}\omega_{s2}} = 1$$

$$\bar{\omega}_{s2} = \frac{(\omega_{s2} - \omega_{s1})\omega_{s2}}{-\omega_{s2}^2 + \omega_{s1}\omega_{s2}} = -1$$

$$\Rightarrow \bar{\omega}_s = 1$$

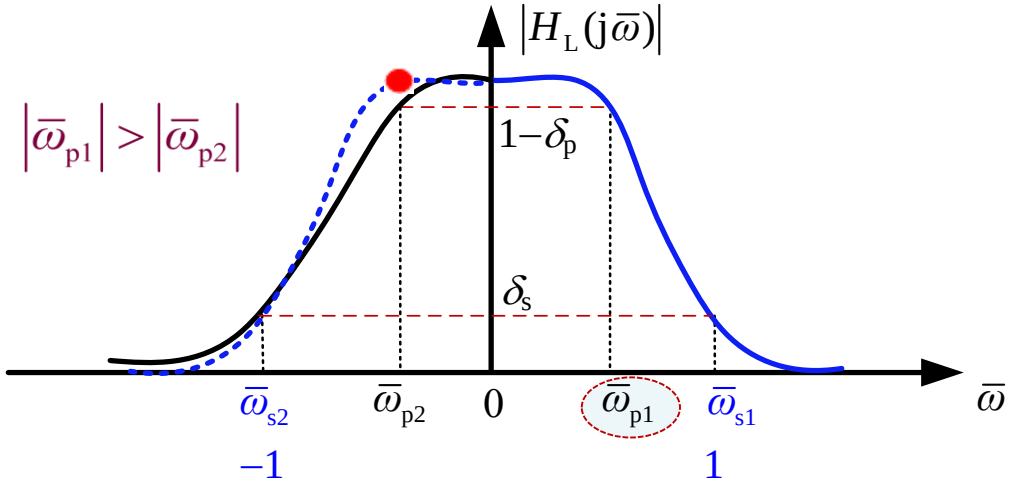
$$\bar{\omega}_{p1} = \frac{B\omega_{p1}}{-\omega_{p1}^2 + \omega_0^2}$$

$$\bar{\omega}_{p2} = \frac{B\omega_{p2}}{-\omega_{p2}^2 + \omega_0^2}$$

$$\Rightarrow \bar{\omega}_p = ?$$

Analog domain frequency conversion

❖ Prototype low pass to band resistance conversion $\bar{\omega} = \dfrac{B\omega}{-\omega^2 + \omega_0^2}$

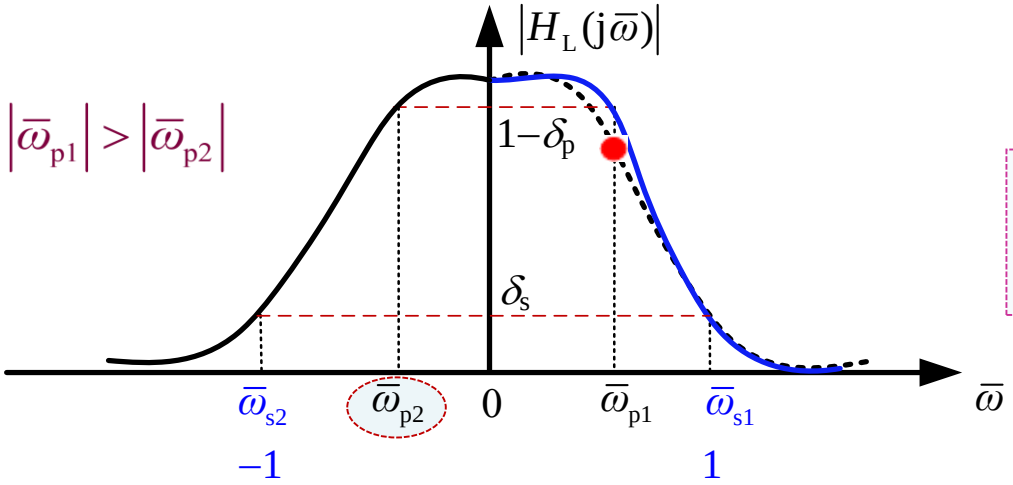


$$\bar{\omega}_p = \max\{|\bar{\omega}_{p1}|,\ |\bar{\omega}_{p2}|\}$$

✓

Analog domain frequency conversion

❖ Prototype low pass to band resistance conversion $\bar{\omega} = \dfrac{B\omega}{-\omega^2 + \omega_0^2}$



$$\bar{\omega}_p = \max\{|\bar{\omega}_{p1}|, |\bar{\omega}_{p2}|\}$$
$$\bar{\omega}_s = |\bar{\omega}_{s1}| = |\bar{\omega}_{s2}| = 1$$

Analog band-stop filter design

❖ Simulate the design steps of a band-stop filter

(1) The parameters in the conversion formula are determined by the stopband intercept frequency of the band-stop filter

$$B = \omega_{s2} - \omega_{s1} \qquad \omega_0^2 = \omega_{s1}\omega_{s2}$$

(2) The on-band and stop-band truncation frequency of prototype low-pass filter is determined($\bar{\omega}_p$ $\bar{\omega}_s$)

$$\bar{\omega}_{p1} = \frac{B\omega_{p1}}{-\omega_{p1}^2 + \omega_0^{\ 2}} \qquad \bar{\omega}_{p2} = \frac{B\omega_{p2}}{-\omega_{p2}^2 + \omega_0^{\ 2}} \qquad \bar{\omega} = \frac{B\omega}{-\omega^2 + \omega_0^2}$$

$$\bar{\omega}_p = \max\{|\bar{\omega}_{p1}|, |\bar{\omega}_{p2}|\}$$

$$\bar{\omega}_s = |\bar{\omega}_{s1}| = |\bar{\omega}_{s2}| = 1$$

Analog band-stop filter design

❖ Simulate the design steps of a band-stop filter

(3) The design passband cut frequency is $\bar{\omega}_p$、Stopband intercept frequency is $\bar{\omega}_s$ =1 (rad/s)、a prototype low-pass filter with passband attenuation of A_p dB and stopband attenuation of A_s dB

(4) Convert prototype low-pass filter to band-stop filter $H_{BS}(s)$

$$H_{BS}(s) = H_L(\bar{s})\Big|_{\bar{s}=\frac{Bs}{s^2+\omega_0^2}}$$

[Example] Try to design a BW analog band-stop filter that meets the following criteria

ω_{p1}=10 rad/s, ω_{p2}=30 rad/s, $A_p \le 1$dB,
ω_{s1}=19 rad/s, ω_{s2}=21 rad/s, $A_s \ge 20$dB

Solution: (1) The parameters in the transformation formula are determined by the stopband intercept frequency of the band-stop filter $B = \omega_{s2} - \omega_{s1} = 2$ $\omega_0^2 = \omega_{s1}\omega_{s2} = 399$

(2) Determine the on-band and stopband truncation frequency of the prototype low-pass filter

$$\bar{\omega}_{p1} = \frac{B\omega_{p1}}{-\omega_{p1}^2 + \omega_0^2} = 0.0669 \quad \bar{\omega}_{p2} = \frac{B\omega_{p2}}{-\omega_{p2}^2 + \omega_0^2} = -0.1198 \qquad \bar{\omega} = \frac{B\omega}{-\omega^2 + \omega_0^2}$$

so $\bar{\omega}_p = \max\{|\bar{\omega}_{p1}|, |\bar{\omega}_{p2}|\} = 0.1198$ $\bar{\omega}_s = 1$

[Example] Try to design a BW analog band-stop filter that meets the following criteria

ω_{p1}=10 rad/s, ω_{p2}=30 rad/s, $A_p \le 1$dB,
ω_{s1}=19 rad/s, ω_{s2}=21 rad/s, $A_s \ge 20$dB

(3) Design a prototype low-pass filter that meets the following criteria

$$\bar{\omega}_p = 0.1198,\ A_p = 1\text{dB},\ \bar{\omega}_s = 1,\ A_s = 20\text{dB}$$

$$N \ge \frac{\lg\left(\dfrac{10^{0.1A_s}-1}{10^{0.1A_p}-1}\right)}{2\lg(\bar{\omega}_s/\bar{\omega}_p)} = 1.4009 \quad \text{Set } N=2 \quad \omega_c = \frac{\bar{\omega}_s}{(10^{0.1A_s}-1)^{1/(2N)}} = 0.3170$$

$$H_L(\bar{s}) = \frac{1}{(\bar{s}/\omega_c)^2 + \sqrt{2}\bar{s}/\omega_c + 1} = \frac{0.1005}{\bar{s}^2 + 0.4483\bar{s} + 0.1005}$$

[Example] Try to design a BW analog band-stop filter that meets the following criteria

ω_{p1}=10 rad/s, ω_{p2}=30 rad/s, $A_p \le 1$dB,
ω_{s1}=19 rad/s, ω_{s2}=21 rad/s, $A_s \ge 20$dB

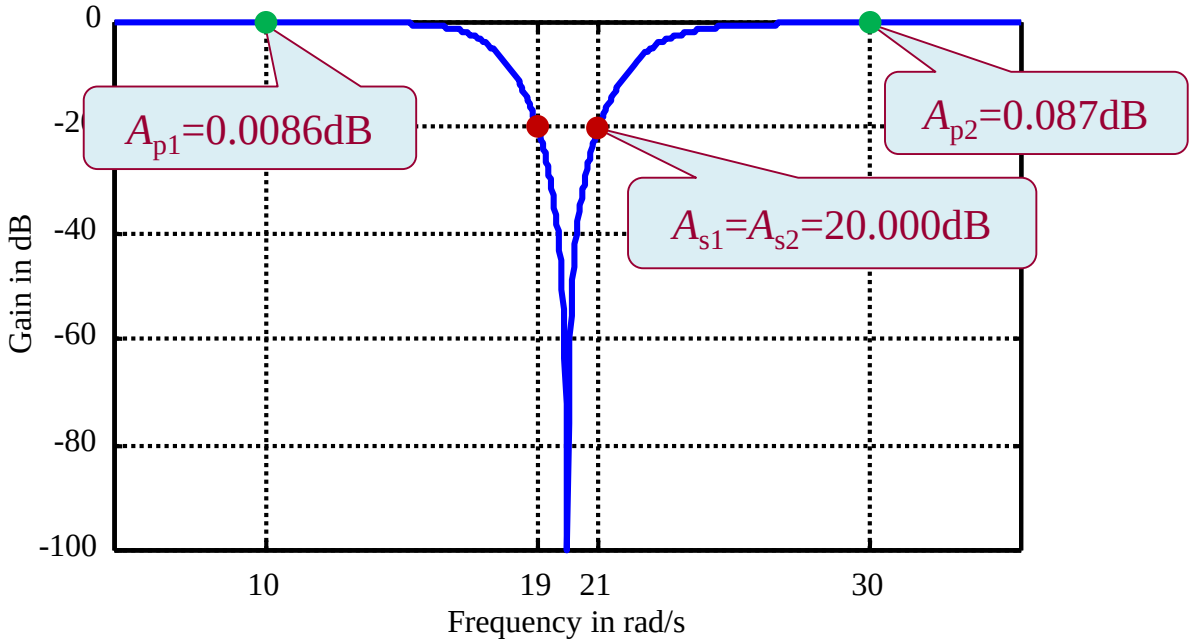
(4) The prototype low-pass filter is converted into a band-stop filter $H_{BS}(s)$

$$H_{BS}(s) = H_L(\bar{s})\Big|_{\bar{s}=\frac{Bs}{s^2+\omega_0^2}}$$

$$= \frac{s^4 + 7.98\times10^2 s^2 + 1.5920\times10^5}{s^4 + 8.9518s^3 + 8.3780\times10^2 s^2 + 3.5718\times10^3 s + 1.5920\times10^5}$$

[Example] Try to design a BW analog band-stop filter that meets the following criteria

ω_{p1}=10 rad/s, ω_{p2}=30 rad/s, $A_p \le$1dB,
ω_{s1}=19 rad/s, ω_{s2}=21 rad/s, $A_s \ge$20dB



Analog domain frequency conversion

Convert the type	Frequency conversion	Complex frequency conversion	Prototype low pass design index	Annotation
Prototype low-pass→high-pass	$\bar{\omega} = 1/\omega$	$\bar{s} = 1/s$	$\bar{\omega}_p = 1/\omega_p$ $\bar{\omega}_s = 1/\omega_s$	$B = \omega_{p2} - \omega_{p1}$
Prototype low-pass→band-pass	$\bar{\omega} = \dfrac{\omega^2 - \omega_0^2}{B\omega}$	$\bar{s} = \dfrac{s^2 + \omega_0^2}{Bs}$	$\bar{\omega}_p = \lvert\omega_{p1}\rvert = \lvert\omega_{p2}\rvert = 1$ $\bar{\omega}_s = \min\{\lvert\bar{\omega}_{s1}\rvert, \lvert\bar{\omega}_{s2}\rvert\}$	$\omega_0^2 = \omega_{p1}\omega_{p2}$ $B = \omega_{s2} - \omega_{s1}$
Prototype low-pass→band elimination	$\bar{\omega} = \dfrac{B\omega}{-\omega^2 + \omega_0^2}$	$\bar{s} = \dfrac{Bs}{s^2 + \omega_0^2}$	$\bar{\omega}_p = \max\{\lvert\bar{\omega}_{p1}\rvert, \lvert\bar{\omega}_{p2}\rvert\}$ $\bar{\omega}_s = \lvert\bar{\omega}_{s1}\rvert = \lvert\bar{\omega}_{s2}\rvert = 1$	$\omega_0^2 = \omega_{s1}\omega_{s2}$

Analog domain frequency conversion

Analog high-pass filter

```
[numt,dent] = lp2hp(num,den,W0)
```

Analog band-pass filter

```
[numt,dent] = lp2bp(num,den,W0,B)
```

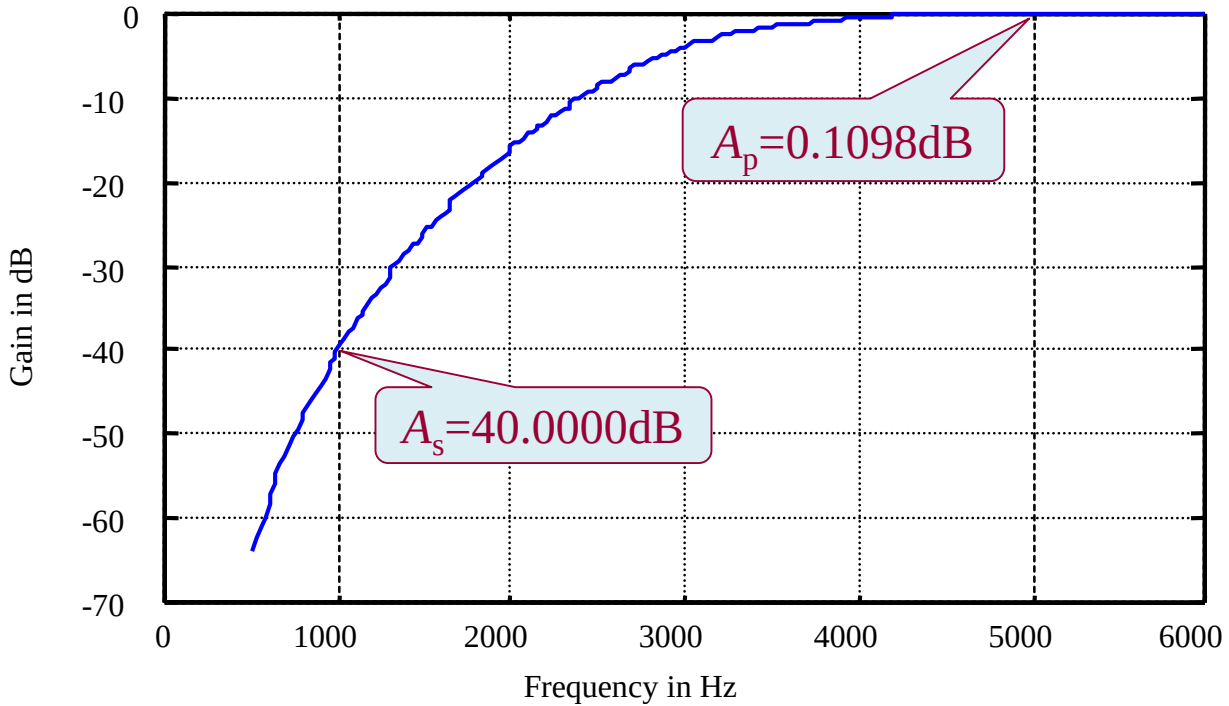
Analog band-stop filter

```
[numt,dent] = lp2bs(num,den,W0,B)
```

[Example] Try to design a BW analog high-pass filter that meets the following criteria f_p=5kHz, f_s=1kHz, $A_p \leq 1$dB, $A_s \geq 40$dB。

```
fp=5000;fs=1000; Ap=1;As=40; % AF specifications
wLp=1/(2*pi*fp); wLs=1/(2*pi*fs); % HP specifications to LP's
[N,Wc]=buttord(wLp,wLs,Ap,As,'s'); % Computer AF order
[num,den] = butter(N,Wc,'s');   % Compute AF coefficients
[numt,dent] = lp2hp(num,den,1); % LP to HP
% display Ap and As
omega=[2*pi*fp, 2*pi*fs];
h = freqs(numt, dent, omega);
fprintf('Ap= %.4f\n',-20*log10(abs(h(1))));
fprintf('As= %.4f\n',-20*log10(abs(h(2))));
% plot the gain
omega = [0: 200: 6000*2*pi];
h = freqs(numt, dent, omega);  % Compute the frequency response of the AF
gain=20*log10(abs(h));  plot(omega/(2*pi),gain);
xlabel('Frequency in Hz');  ylabel('Gain in dB');
```

[Example] Try to design a BW analog high-pass filter that meets the following criteria f_p=5kHz, f_s=1kHz, $A_p \leq$1dB, $A_s \geq$40dB。



[Example] Try to design a BW analog bandpass filter that meets the following criteria ω_{p1}=6 rad/s, ω_{p2}=8 rad/s, $A_p \le$1dB,

ω_{s1}=4 rad/s, ω_{s2}=11 rad/s, $A_s \ge$32dB

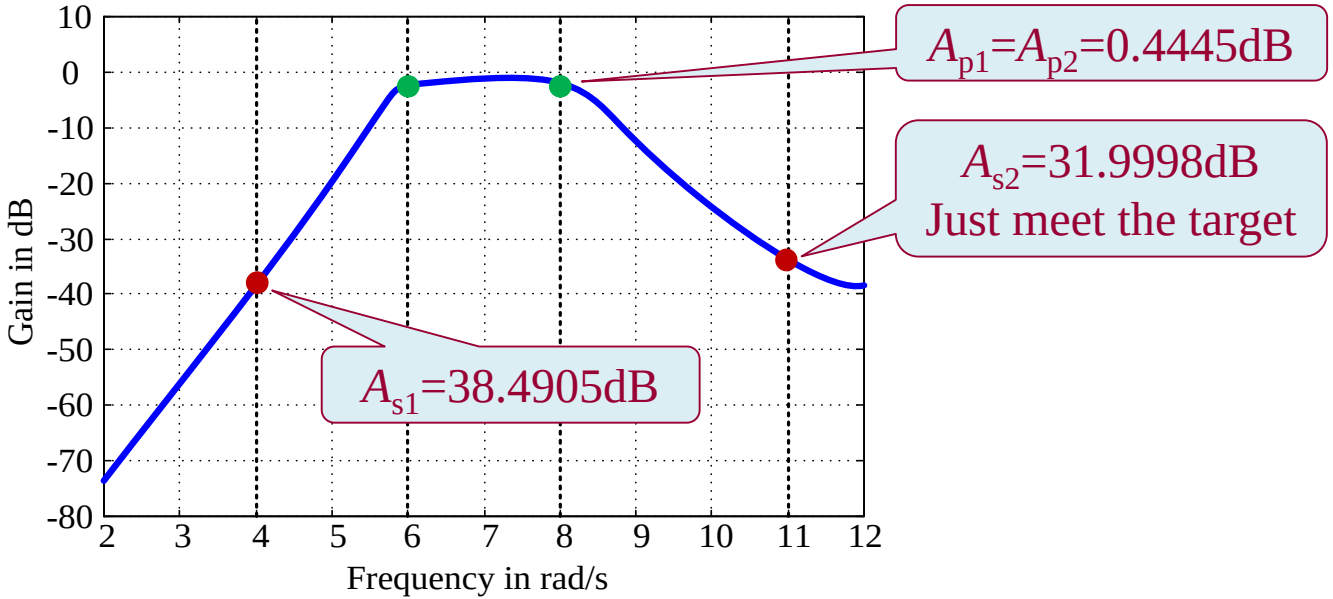
```
wp=1; ws=3.3182;  Ap=1;  As=32;   % AF low-pass specifications
w0=sqrt(48); B=2;
[N,Wc]=buttord(wp,ws,Ap,As,'s');  % Computer low-pass AF order
[num,den] = butter(N,Wc,'s'); % Compute  low-pass AF coefficients
[numt,dent] = lp2bp(num,den,w0,B); % LP to BP
omega=[6 8 4 11];
h = freqs(numt,dent,omega); %Compute Ap and As of the AF
fprintf('Ap1= %.4f\n',-20*log10(abs(h(1))));
fprintf('Ap2= %.4f\n',-20*log10(abs(h(2))));
fprintf('As1= %.4f\n',-20*log10(abs(h(3))));
fprintf('As2= %.4f\n',-20*log10(abs(h(4))));
w=linspace(2,12,1000);
h=freqs(numt,dent,w);
plot(w,20*log10(abs(h))) ; grid on;
xlabel('Frequency in rad/s');
ylabel('Gain in dB')
```

$\bar{\omega}_{s1} = -4, \quad \bar{\omega}_{s2} = 3.3182$

$\bar{\omega}_s = \min\{|\bar{\omega}_{s1}|, |\bar{\omega}_{s2}|\} = 3.3182$

[Example] Try to design a BW analog bandpass filter that meets the following criteria ω_{p1}=6 rad/s, ω_{p2}=8 rad/s, $A_p \le$1dB,

ω_{s1}=4 rad/s, ω_{s2}=11 rad/s, $A_s \ge$32dB



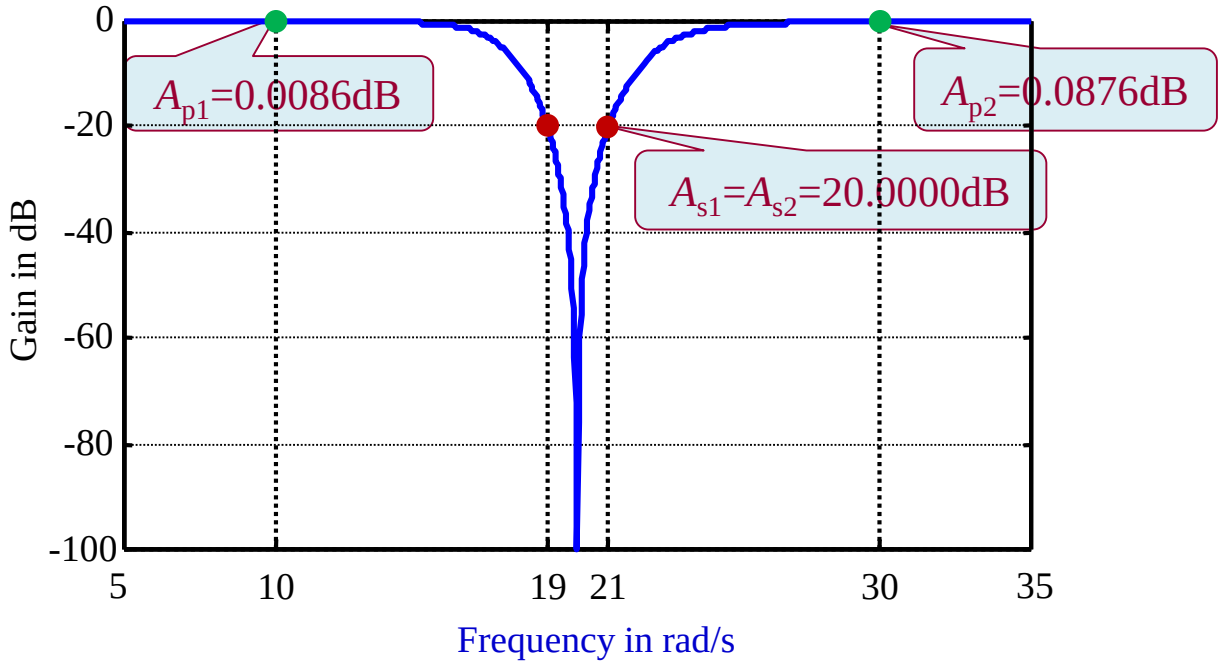
[Example] Try to design a BW analog band-stop filter that meets the following criteria ω_{p1}=10 rad/s, ω_{p2}=30 rad/s, $A_p \le 1$dB,

ω_{s1}=19 rad/s, ω_{s2}=21 rad/s, $A_s \ge 20$dB

```
Ap=1;As=20; wp1=10; wp2=30; ws1=19; ws2=21; % BS AF specification
B=ws2-ws1;w0=sqrt(ws1*ws2);
wLp1=B*wp1/(w0*w0-wp1*wp1); wLp2=B*wp2/(w0*w0-wp2*wp2);
wLp=max(abs(wLp1),abs(wLp2)); wLs=1;
[N,Wc]=buttord(wLp, wLs, Ap, As, 's' ); % Computer low-pass AF order
[num,den] = butter(N, Wc, 's' );   % Compute low-pass AF coefficients
[numt,dent]=lp2bs(num,den,w0,B);  % LP to BS
omega=[10 30 19 21];
h = freqs(numt,dent,omega);
fprintf('Ap1= %.4f\n',-20*log10(abs(h(1))));
fprintf('Ap2= %.4f\n',-20*log10(abs(h(2))));
fprintf('As1= %.4f\n',-20*log10(abs(h(3))));
fprintf('As2= %.4f\n',-20*log10(abs(h(4))));
w=linspace(5,35,1000);  h=freqs(numt,dent,w);
plot(w,20*log10(abs(h)));
xlabel('Frequency in rad/s');  ylabel('Gain in dB')
```

[Example] Try to design a BW analog band-stop filter that meets the following criteria ω_{p1}=10 rad/s, ω_{p2}=30 rad/s, $A_p \le 1$dB,

ω_{s1}=19 rad/s, ω_{s2}=21 rad/s, $A_s \ge 20$dB



Thank you

Some of the materials quoted in this course come from many years of teaching accumulation of the lecturer, from various media and the communication of colleagues and peers, it is difficult to specify the sources.